Task Profile Definition

Each task type τ_i is characterized by a static prior profile, refined at runtime via exponential moving average (EMA):

$$\mathcal{P}(\tau_i) = \langle v_i,\ d_i,\ s_i,\ \iota_i,\ r_i,\ c_i \rangle \tag{1}$$

where:

- v_i — estimated peak GPU VRAM usage (MB)
- $d_i \in \{\text{gpu_pinned},\ \text{gpu_preferred},\ \text{cpu_preferred}\}$ — device affinity class
- $s_i \in [0, 1]$ — GPU sensitivity (fraction of performance gained from GPU)
- $\iota_i \in [0, 1]$ — interactivity (user-visible urgency)
- r_i — CPU-to-GPU execution time ratio ($T_{\text{cpu}}/T_{\text{gpu}}$)
- $c_i \in \{0, 1\}$ — whether a CPU execution path exists

Hybrid Affinity Profiling

The effective CPU-to-GPU ratio is a blend of static prior and runtime EMA:

$$\hat{r}_i = \begin{cases} r_i & \text{if no runtime samples} \\ \alpha \cdot r_i^{\text{obs}} + (1 - \alpha) \cdot r_i & \text{otherwise} \end{cases} \tag{2}$$

where $\alpha = 0.3$ is the EMA smoothing factor and r_i^{obs} is the observed ratio from per-task-type timing measurements.

Task Classification (Four Tiers)

1. **GPU-pinned heavy burst** ($d_i = \text{gpu_pinned}$): Image generation. Must use GPU; forces asymmetric migration of LLM.
2. **GPU-preferred migratable burst** ($d_i = \text{gpu_preferred}$): Neural TTS (F5-TTS, Qwen-TTS). Benefits from GPU but can degrade to CPU.
3. **CPU-affine auxiliary stage** ($d_i = \text{cpu_preferred}$): OCR, RAG embedding (BGE-small), document parsing. Lightweight; GPU overhead exceeds benefit.
4. **Resident migratable service**: LLM (Qwen-2.5-7B). Resides on GPU; offloaded to CPU under contention.

Cost-Benefit Routing Decision

Core principle: EdgeSched does not route a task to GPU merely because GPU is available; it routes a task to GPU *only when the expected acceleration outweighs the disruption to resident GPU state.*

GPU Benefit

$$B_{\text{gpu}}(\tau_i) = \big(T_{\text{cpu}}(\tau_i) - T_{\text{gpu}}(\tau_i)\big) \cdot s_i \tag{3}$$

$$T_{\text{cpu}}(\tau_i) = T_{\text{gpu}}(\tau_i) \cdot \hat{r}_i \tag{4}$$

where $T_{\text{gpu}}(\tau_i)$ is the EMA-corrected GPU execution time and s_i is the GPU sensitivity of task τ_i.

Disruption Cost

$$C_{\text{disrupt}} = \begin{cases} T_{\text{offload}} + T_{\text{restore}} + P_{\text{degrade}} & \text{if LLM resides on GPU} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where T_{offload} and T_{restore} are the measured (non-symmetric) migration durations and P_{degrade} is the UX degradation penalty while the LLM runs on CPU.

Routing Rule

$$\text{route}(\tau_i) = \begin{cases} \text{CPU} & \text{if } d_i = \text{cpu_preferred} \\ \text{GPU} & \text{if } d_i = \text{gpu_pinned} \\ \text{CPU} & \text{if } c_i = 1 \ \wedge \ B_{\text{gpu}}(\tau_i) < C_{\text{disrupt}} \\ \text{GPU} & \text{otherwise} \end{cases} \tag{6}$$

Admission Decision (given GPU routing)

Once routed to GPU, the admission controller decides whether to execute immediately (with migration), queue, or reject:

$$C_{\text{migrate}} = T_{\text{offload}} + T_{\text{burst}} + T_{\text{restore}} + P_{\text{degrade}} \tag{7}$$

$$C_{\text{queue}} = q \cdot T_{\text{burst}} + T_{\text{burst}} \tag{8}$$

$$\text{decision} = \begin{cases} \text{ADMIT_NOW} & \text{if } C_{\text{migrate}} < C_{\text{queue}} \\ \text{QUEUE} & \text{if } q \cdot T_{\text{burst}} < W_{\max} \\ \text{REJECT} & \text{otherwise} \end{cases} \tag{9}$$

where q is the current queue depth and $W_{\max}$ is the maximum acceptable queueing delay.